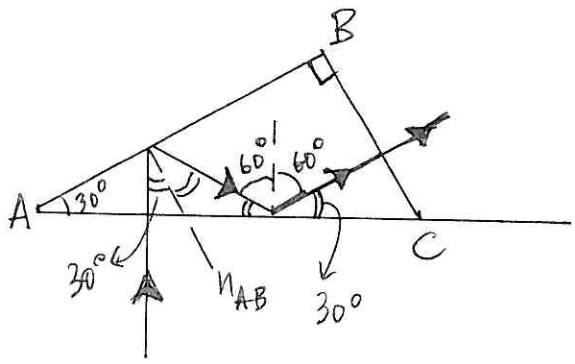
HW01 $\boxed{P1}$



$n = 2.1$

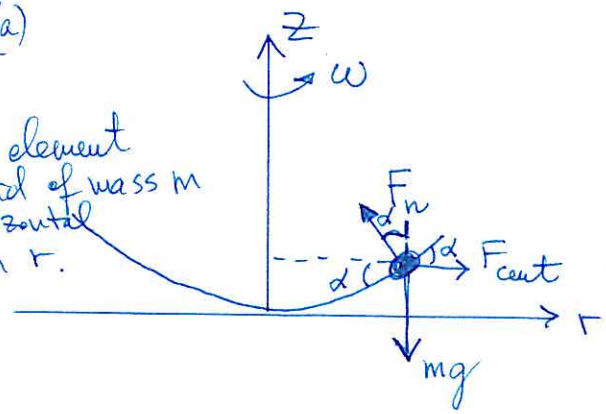
It is seen that light strikes AB at 30° with respect to normal n_{AB}. It then strikes AC at 60° and is reflected through BC side.

Since $n = 2.1$, the angle of total internal reflection for the prism is $\theta_{crit} = \sin^{-1}\left(\frac{1}{2.1}\right) \cong 28°$.

Hence no light is coming out of AB and AC sides $\Rightarrow$ the answer is (B).

$\boxed{P2}$ (a)

Consider element of liquid of mass m at horizontal location r.



$\vec{F}_n$ = normal to surface support force acting on element of mass m from all other elements

$\vec{F}_{cent}$ = centrifugal force

$$\left.\begin{aligned} F_{cent} &= m\omega^2 r = F_n \sin\alpha \\ mg &= F_n \cos\alpha \end{aligned}\right\} \Rightarrow \tan(\alpha) = \frac{m\omega^2 r}{mg} = \frac{\omega^2 r}{g}$$

The shape of liquid $z(r)$ is found from

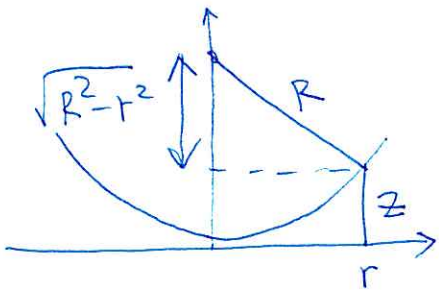
$$\frac{dz}{dr} = \tan(\alpha) = \frac{\omega^2 r}{g} \quad \Rightarrow \text{integrate both sides over } r$$

$$\Rightarrow z(r) - z(0) = \int_0^r \frac{\omega^2 r}{g}\, dr = \frac{\omega^2 r^2}{2g}$$

$$z(r) = \frac{\omega^2 r^2}{2g} + c \quad , \text{where } z(r=0) = c.$$

The shape is parabolic.

(b)



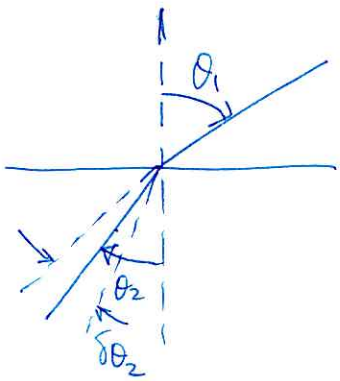
For the sphere $z = R - \sqrt{R^2 - r^2} \approx \frac{r^2}{2R}$

$\Rightarrow$ hence the figure of rotation obtained in (a) can be approximated as a spherical convex mirror of radius $R = \frac{g}{\omega^2}$.

the focal length is $f = \frac{1}{2}R = \frac{g}{2\omega^2}$.

$$\Rightarrow \omega = \sqrt{\frac{g}{2f}} \approx \sqrt{\frac{10}{2 \cdot 0.1}} = \sqrt{50} \approx 7 \text{ rad/s}$$

P3



Snell's law

$\sin\theta_1 = n(\lambda)\sin\theta_2(\lambda)$

where $\theta_2(\lambda)$ denotes refraction angle of the wave with wavelength λ.

Differentiate both sides by λ:

$\frac{d}{d\lambda}\sin\theta_1 = 0$ because incident angle is the same for all λ.

$$\Rightarrow \frac{d}{d\lambda}\left(n(\lambda)\sin\theta_2(\lambda)\right) = 0$$

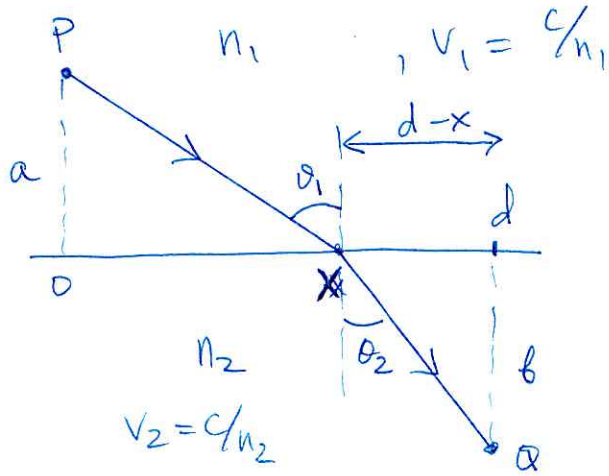
$$\frac{dn}{d\lambda}\sin\theta_2 + n\cos\theta_2\frac{d\theta_2}{d\lambda} = 0$$

$$\frac{d\theta_2}{d\lambda} = -\frac{\sin\theta_2}{\cos\theta_2}\frac{1}{n}\frac{dn}{d\lambda}$$

$$\Rightarrow \delta\theta_2 = \left|\frac{d\theta_2}{d\lambda}\right|\delta\lambda = \left|\tan(\theta_2)\frac{1}{n}\frac{dn}{d\lambda}\right|\delta\lambda$$

the answer is (E).

P4



(a) $t_{PQ} = t_{PX} + t_{XQ}$

$$t_{PX} = \frac{\sqrt{x^2+a^2}}{c/n_1} \quad , \quad t_{XQ} = \frac{\sqrt{b^2+(d-x)^2}}{c/n_2}$$

(b) Require $\dfrac{dt_{PQ}}{dx} = 0 = \dfrac{n_1}{c}\underbrace{\dfrac{x}{\sqrt{a^2+x^2}}}_{\sin\theta_1} - \dfrac{n_2}{c}\underbrace{\dfrac{d-x}{\sqrt{b^2+(d-x)^2}}}_{\sin\theta_2}$

$$\Rightarrow \quad n_1 \sin\theta_1 = n_2 \sin\theta_2 \quad \text{(Snell's law!)}$$